

Assignment - 1

Q1. To Prove: $n^2 < 2^n \quad \forall n \geq 5, n \in \mathbb{N}$

For $n=5$, $5^2 < 2^5 \Rightarrow 25 < 32$ [True]

For $n=k$, let $k^2 < 2^k$ be true $[k \in \mathbb{N}, k \geq 5]$ ①

For $n=k+1$, $(k+1)^2 < 2^{k+1}$ [Assumed True] ~~⊕~~

$$\Rightarrow \cancel{k^2 + 2k + 1} < \cancel{2 \cdot 2^k}$$

~~Now, for any $k \geq 1$, $2k \geq 2 \Rightarrow 2k+1 > 2$~~

~~But, from ①, $k^2 < 2^k \Rightarrow 2k^2 < 2 \cdot 2^k$~~

$$\therefore \cancel{k^2 + 2k + 1} < \cancel{2 \cdot 2^k}$$

$$\text{Now, } (k+1)^2 = k^2 + 2k + 1 < k^2 + 2k + 1 + \underbrace{2k + 4}_{(>0 : k \geq 5)}$$

$$\Rightarrow (k+1)^2 < k^2 + 4k + 5 < k^2 + 4k + \underbrace{k - 5}_{(>0 : k \geq 5)} + 5$$

$$\Rightarrow (k+1)^2 < k^2 + 5k < k^2 + 5k + \underbrace{k(k-5)}_{(>0 : k \geq 5)}$$

$$\Rightarrow (k+1)^2 < 2k^2 + 5k - 5k$$

$$\text{But, from ①, } k^2 < 2^k \Rightarrow 2k^2 < 2 \cdot 2^k$$

$$\Rightarrow (k+1)^2 < 2^{k+1}, \text{ which is now proved to be true.}$$

$\therefore$ By PMI, $n^2 < 2^n$ holds $\forall n \geq 5, n \in \mathbb{N}$.

Q2. Let ~~not~~ $\sup A$ be supremum of A . (2)

$$\Rightarrow \sup A \geq a \quad \forall a \in A$$

$$\Rightarrow -\sup A \leq -a \quad - (1)$$

Also, ~~$\forall b \in B$ s.t. $b =$~~

Also, $\exists a \in A$ s.t. $b = -a$, $b \in B$.

(by def. of B)

$$\text{From (1), } b \geq -\sup A \quad \forall b \in B \quad - (2)$$

$$\text{But, } \inf B \leq b \quad \forall b \in B \quad - (3)$$

(where $\inf B$ is infimum of B).

$$\therefore \inf B = -\sup A \quad (2 \& 3)$$

Also, since $\inf B$ exists, B is bounded below.

Q3. a) Consider the following set of the form p/q .

$n=1$	$n=2$	$n=3$	$n=4$
$\pm 1/1$	$\pm 2/1$	$\pm 3/1$	$\pm 4/1$
$\pm 1/2$	$\pm 2/2$	$\pm 3/2$	$\pm 3/3$
$\pm 1/3$	$\pm 2/3$	$\pm 3/3$	$\pm 4/3$
$\pm 1/4$	$\pm 2/4$	$\pm 3/4$	$\pm 4/4$

Consider a diagonal d_n s.t. $p+q=n$.

$\Rightarrow$ Any p/q will have d_n of finite length

Our bijection is just to go along the diagonal 1, 2, 3, 4. that count all $\pm p/q$ that are rational (skip any p/q with $\text{HCF}(p,q) \neq 1$)

$$f(1) = 0$$

$$f(2) = 1$$

$$f(3) = -1$$

$$f(4) = 1/2$$

$$f(5) = -1/2$$

... and so on

in this way we can cover all rational numbers in terms of $f(n) \quad \forall n \in \mathbb{N}$.

$\therefore$ There exists a bijection $f: \mathbb{N} \rightarrow \mathbb{Q}$.

$\Rightarrow N \sim Q \Rightarrow Q$ is countable.

(3)

f is injective since any p/q with $\text{HCF}(p, q) \neq 1$ was rejected, giving unique rational no.s.

Also, f is surjective as every rational no. appears somewhere in the grid.

$\therefore f$ is bijective $\Rightarrow Q \sim N \Rightarrow Q$ is countable

Now, $A \subseteq Q$ $[A = \{x \in Q \mid -100 \leq x \leq 100\}]$,

and every subset of a countable set is countable, so:

A is countable

b) $B = \{(x, y) : x \in N, y \in Z\}$

$\Rightarrow B = N \times Z$ (def. of cartesian product)

It is true that the cartesian product of two countable sets yields a countable set.

$\therefore$ B is countable

c) Let $C = [0, 0.1]$ be countable.

$\Rightarrow$ There exists a bijection b/w C & N .

Consider a bijection $f: N \rightarrow C$ s.t.

$$f(1) = 0.0a_{11}a_{12} \dots a_{1n}$$

$$f(2) = 0.0a_{21}a_{22} \dots a_{2n}$$

$\vdots$

$$f(n) = 0.0a_{n1}a_{n2} \dots a_{nn}$$

Let $p \in C \Rightarrow p = 0.p_1p_2p_3 \dots p_n$

$\Rightarrow p_1 \in [1, 9] \cap N$. Let $p_1 \neq a_{11}$

$p_2 \in [1, 9] \cap N$. Let $p_2 \neq a_{22}$

$p_n \in [1, 9] \cap N$. Let $p_n \neq a_{nn}$

4
 $\Rightarrow p$ is different from every listed no.
by at least one digit.

$\therefore p$ is not counted in f .

$\therefore [0, 0.1] = \mathbb{C}$ is uncountable.

2) $D = \left\{ \frac{1}{n} \mid n \in \mathbb{N} \right\}$

Consider a function $f: \mathbb{N} \rightarrow D$,

s.t. $f(n) = \frac{1}{n} \quad \forall n \in \mathbb{N} \text{ \& } \frac{1}{n} \in D$.

Let $n_1, n_2 \in \mathbb{N}$. If $f(n_1) = f(n_2)$

$$\Rightarrow \frac{1}{n_1} = \frac{1}{n_2} \Rightarrow \underline{n_1 = n_2}$$

$\therefore f$ is injective.

①

Now, let $m = \frac{1}{n} \Rightarrow n = \frac{1}{m}$

$$\Rightarrow f(n) = \frac{1}{n} = \frac{1}{1/m} = m$$

$\Rightarrow \forall m \in \frac{1}{n}, \exists n = \frac{1}{m}$ s.t. $f(n) = m$.

$\therefore f$ is surjective.

②

From ① & ②, f is bijective.

$\Rightarrow D \sim \mathbb{N} \Rightarrow \underline{D \text{ is countable.}}$

Q4. Consider $A = (0, 1)$.

(5)

If even one subset is uncountable, the set is uncountable. Clearly, $A \subset \mathbb{R}$

Let A be countable. ~~early~~

$\Rightarrow \exists f: \mathbb{N} \rightarrow A$ where f is bijective.

The following grid can exist because all real nos. have decimal expansions.

$$f(1) = 0. a_{11} a_{12} \dots a_{1n}$$

$$f(2) = 0. a_{21} a_{22} \dots a_{2n}$$

$$f(n) = 0. a_{n1} a_{n2} \dots a_{nn}$$

Consider $x \in \mathbb{R} \cap A$ s.t.

$$x = 0. x_1 x_2 \dots x_n$$

where, $x_i \neq a_{ii}$, $x_i \in [1, 9] \cap \mathbb{N}$.

$$\Rightarrow c \in f(\mathbb{N})$$

However, c is not among $f(1), f(2), \dots, f(n)$.

$$\Rightarrow c \notin f(\mathbb{N})$$

Thus, the contradiction arose due to the false assumption that f is a bijection, that is, A is countable.

$\Rightarrow A$ is uncountable (by contradiction).

Also, $A = (0, 1) \subset \mathbb{R}$.

$\therefore \boxed{\mathbb{R} \text{ is uncountable}}.$

Q5.

(6)

a) Let $x \in f^{-1}\left(\bigcup_{i \in I} B_i\right)$

$$\Rightarrow f(x) \in \bigcup_{i \in I} B_i$$

$$\Rightarrow f(x) \in B_1 \text{ or } f(x) \in B_2 \text{ or } \dots \text{ or } f(x) \in B_n$$

$$\Rightarrow x \in f^{-1}(B_1) \text{ or } x \in f^{-1}(B_2) \text{ or } \dots$$

$$\Rightarrow x \in f^{-1}(B_1) \cup f^{-1}(B_2) \cup \dots$$

$$\Rightarrow x \in \bigcup_{i \in I} f^{-1}(B_i)$$

$$\therefore f^{-1}\left(\bigcup_{i \in I} B_i\right) \subseteq \bigcup_{i \in I} f^{-1}(B_i) \quad - (1)$$

Let $y \in \bigcup_{i \in I} f^{-1}(B_i)$

$$\Rightarrow y \in f^{-1}(B_1) \text{ or } y \in f^{-1}(B_2) \text{ or } \dots$$

$$\Rightarrow f(y) \in B_1 \text{ or } f(y) \in B_2 \text{ or } \dots$$

$$\Rightarrow f(y) \in B_1 \cup B_2 \cup \dots$$

$$\Rightarrow f(y) \in \bigcup_{i \in I} B_i$$

$$\Rightarrow y \in f^{-1}\left(\bigcup_{i \in I} B_i\right)$$

$$\therefore \bigcup_{i \in I} f^{-1}(B_i) \subseteq f^{-1}\left(\bigcup_{i \in I} B_i\right) \quad - (2)$$

From (1) & (2) :

$$f^{-1}\left(\bigcup_{i \in I} B_i\right) = \bigcup_{i \in I} f^{-1}(B_i)$$

b) Let $x \in f^{-1}\left(\bigcap_{i \in I} B_i\right)$

(7)

$$\Rightarrow f(x) \in \bigcap_{i \in I} B_i$$

$$\Rightarrow f(x) \in B_1 \text{ and } f(x) \in B_2 \text{ and } \dots$$

$$\Rightarrow x \in f^{-1}(B_1) \text{ and } x \in f^{-1}(B_2) \text{ and } \dots$$

$$\Rightarrow x \in \bigcap_{i \in I} f^{-1}(B_i)$$

$$\therefore f^{-1}\left(\bigcap_{i \in I} B_i\right) \subseteq \bigcap_{i \in I} f^{-1}(B_i) \quad - (1)$$

Let $y \in \bigcap_{i \in I} f^{-1}(B_i)$

$$\Rightarrow y \in f^{-1}(B_1) \text{ and } y \in f^{-1}(B_2) \text{ and } \dots$$

$$\Rightarrow f(y) \in B_1 \text{ and } f(y) \in B_2 \text{ and } \dots$$

$$\Rightarrow f(y) \in \bigcap_{i \in I} B_i$$

$$\Rightarrow y \in f^{-1}\left(\bigcap_{i \in I} B_i\right)$$

$$\therefore \bigcap_{i \in I} f^{-1}(B_i) \subseteq f^{-1}\left(\bigcap_{i \in I} B_i\right) \quad - (2)$$

From (1) & (2):

$$f^{-1}\left(\bigcap_{i \in I} B_i\right) = \bigcap_{i \in I} f^{-1}(B_i)$$

Q8. To Prove :

$$\forall \epsilon > 0 \exists n \in \mathbb{N} \text{ s.t.}$$

$$n \geq N \Rightarrow 2 - \epsilon < \frac{2n+1}{n+2} < 2 + \epsilon$$

Proof:

$$\begin{aligned} \frac{2n+1}{n+2} &= \frac{2n+4-3}{n+2} = \frac{2(n+2)}{n+2} - \frac{3}{n+2} \\ &= 2 - \frac{3}{n+2} \end{aligned}$$

$$\forall \epsilon > 0, \quad \frac{2n+1}{n+2} = 2 - \frac{3}{n+2} < 2 + \epsilon$$

-(1)

By Archimedian property,

$$\forall \epsilon > 0 \exists n \in \mathbb{N} \text{ s.t.}$$

$$\epsilon > \frac{1}{n}$$

$$\text{Let } \epsilon' = 3\epsilon$$

$$\Rightarrow \epsilon = \epsilon'/3$$

$$\Rightarrow \frac{\epsilon'}{3} > \frac{1}{n}$$

$$\Rightarrow \epsilon' > \frac{3}{n}$$

$$\Rightarrow \epsilon' > \frac{3}{n+2} \quad \left[\because \forall n \in \mathbb{N}, \frac{1}{n} < \frac{1}{n+2} \right]$$

$$\Rightarrow -\epsilon' < -\frac{3}{n+2}$$

$$\Rightarrow 2 - \epsilon' < 2 - \frac{3}{n+2} = \frac{2(n+2) - 3}{n+2}$$

$$\Rightarrow 2 - \epsilon' < \frac{2n+1}{n+2} \quad \text{--- (2)}$$

From (1) & (2),

(9)

$$2 - \varepsilon < \frac{2n+1}{n+2} < 2 + \varepsilon$$

$\forall \varepsilon > 0, n \in \mathbb{N}$

Hence Proved.

Q7. $P(n): 1^2 + 2^2 + \dots + n^2 = \frac{2n^3 + 3n^2 + n}{6}$

$$P(1): 1^2 = \frac{2(1)^3 + 3(1)^2 + 1}{6} = \frac{6}{6} = 1$$

$\Rightarrow P(1)$ is true.

$$P(k): 1^2 + 2^2 + \dots + k^2 = \frac{2k^3 + 3k^2 + k}{6}$$

$[k \in \mathbb{N}]$

Let $P(k)$ be true.

$$\Rightarrow P(k+1): 1^2 + 2^2 + \dots + k^2 + (k+1)^2 = \frac{2(k+1)^3 + 3(k+1)^2 + (k+1)}{6}$$

$$\Rightarrow \frac{2k^3 + 3k^2 + k}{6} + (k+1)^2 = \frac{2(k+1)^3 + 3(k+1)^2 + (k+1)}{6}$$

(from $P(k)$)

$$\Rightarrow \cancel{2k^3 + 3k^2 + k} + 6k^2 + 12k + 6$$

$$= \cancel{2k^3} + \cancel{2} + \cancel{6k^2} + \cancel{6k} + 3k^2 + 8k + 5 + \cancel{k+1}$$

true

$$\Rightarrow 0 = 0 \Rightarrow P(k+1) \text{ is true given } P(k) \text{ is true}$$

$\forall k \in \mathbb{N}$.

$\therefore$ By PMI, $P(n)$ is true $\forall n \in \mathbb{N}$

$$\Rightarrow 1^2 + 2^2 + \dots + n^2 = \frac{2n^3 + 3n^2 + n}{6}, \quad n \geq 1$$

Q8. $P(n) : 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} < 2\sqrt{n-1}$

$\forall n \geq 2, n \in \mathbb{N}.$

$P(2) : 1 + \frac{1}{\sqrt{2}} < 2\sqrt{2-1}$

$\Rightarrow 1 + \frac{1}{\sqrt{2}} < 2 \Rightarrow \frac{1}{\sqrt{2}} < 1$

$\Rightarrow \sqrt{2} > 1$, which is true.

$\Rightarrow P(2)$ is true.

$P(k) : 1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} < 2\sqrt{k-1}$ — (1)
 $[k \in \mathbb{N}, k \geq 2]$

Let $P(k)$ be true.

$P(k+1) : 1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k+1}} < 2\sqrt{k+1-1}$

$\Rightarrow P(k) + \frac{1}{\sqrt{k+1}} < 2\sqrt{k}$

From (1), $1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} < 2\sqrt{k-1} + \frac{1}{\sqrt{k+1}}$

$\Rightarrow P(k+1) < 2\sqrt{k^2-k} + \frac{1}{\sqrt{k+1}}$ — (2)

Now, $\left(k - \frac{1}{2}\right)^2 = k^2 - k + \frac{1}{4}$

$\Rightarrow \left(k - \frac{1}{2}\right)^2 > k^2 - k$

$\Rightarrow k - \frac{1}{2} > \sqrt{k^2 - k}$ — (3)

② & ③:-

①

$$P(k+1) < \frac{2(k-\frac{1}{2})+1}{\sqrt{k}}$$

$$P(k+1) < \frac{2k-1+1}{\sqrt{k}}$$

$$\Rightarrow P(k+1) < 2\sqrt{k}$$

$\therefore P(k+1)$ is also true.

H.P.

Q9.

$$a_1 = \sqrt{2}$$

$$a_2 = \sqrt{2+\sqrt{2}}$$

(12)

$$P(1) : a_1 = \sqrt{2}$$

$$\Rightarrow 1 < a_1 < 2$$

$\Rightarrow P(1)$ is true.

$\Rightarrow P(n)$ is also true.

$$\Rightarrow 1 < a_n < 2$$

$$\Rightarrow \sqrt{1} < \sqrt{a_n} < \sqrt{2}$$

$$\Rightarrow 1+2 < \sqrt{a_n} + 2 < 2+\sqrt{2}$$

$$\Rightarrow \sqrt{3} < \sqrt{2+\sqrt{a_n}} < \sqrt{2+\sqrt{2}}$$

$$\Rightarrow 1.73 < \sqrt{2+\sqrt{a_n}} < 1.84$$

$$\Rightarrow 1.73 < a_{n+1} < 1.84$$

$$\Rightarrow 1 < a_{n+1} < 2$$

H.P.

(By PMI)

$[P(n) \text{ is true } \forall n]$